

# Engineering Arena 2026

Complete Solution Set — Helping Stemo Rescue Athena

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## Abstract

This document contains the worked solutions, governing equations and calculations for all 18 interactive tasks of Engineering Arena 2026. For each task the engineering reasoning is shown, followed by an **orange box** giving the exact value to type or the action to perform in the interactive web page. Tasks whose on-screen data is hidden (e.g. an offline scanner) are solved symbolically, with the procedure to apply once the values are revealed.

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## 1 Task 1 — Column Stability — Euler Buckling of a Braced Column

**Problem.** A pinned escape-hatch column of length  $L = 4.0$  m and flexural rigidity  $EI = 80$  kN·m<sup>2</sup> carries an axial load  $P = 56.0$  kN. A movable brace at height  $a$  splits the column into two segments. The critical load is governed by the *longer* unsupported segment. We must position the brace so the column does not buckle.

### Governing relations

$$P_{cr} = \frac{\pi^2 EI}{L_e^2}, \quad L_e = \max(a, L - a), \quad \text{Win: } P_{cr} \geq P_{applied}.$$

**Solution.** Rearranging the buckling condition for the maximum admissible effective length:

$$P_{cr} \geq P \implies L_e \leq \pi \sqrt{\frac{EI}{P}} = \pi \sqrt{\frac{80}{56}} = 3.755 \text{ m.}$$

Since  $L_e = \max(a, L - a)$ , the brace must satisfy  $a \leq 3.755$  and  $L - a \leq 3.755$ , i.e.  $0.245 \leq a \leq 3.755$  m. The *strongest* configuration places the brace at mid-height,  $a = L/2 = 2.0$  m, which gives  $L_e = 2.0$  m and

$$P_{cr} = \frac{\pi^2(80)}{2.0^2} = 197.4 \text{ kN}, \quad \text{SF} = \frac{197.4}{56} = 3.53.$$

This is the “brace at midpoint  $\rightarrow P_{cr} \times 4$ ” key insight: halving  $L_e$  quadruples  $P_{cr}$ .

### INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW

Drag the brace to mid-height  $a = 2.00$  m (midpoint).

Result:  $L_e = 2.00$  m,  $P_{cr} = 197.4$  kN  $\geq 56$  kN. Press **LOCK BRACE**.

(Any  $a$  in 0.245–3.755 m satisfies the win condition;  $a = 2.0$  m maximises  $P_{cr}$ .)

## 2 Task 2 — Solar Sail MPPT — Maximum Power Point Tracking

**Problem.** A research-satellite solar panel operates on a non-linear  $P$ – $V$  curve. The Maximum Power Point Tracker must sit at the voltage that maximises output power. Sail  $\alpha$ :  $I_{sc} = 8$  A,  $V_{oc} = 20$  V, irradiance  $1000$  W/m<sup>2</sup>.

### Governing relations

$$P(V) = V \cdot I(V), \quad \text{MPP: } \left. \frac{dP}{dV} \right|_{V=V_{mp}} = 0.$$

$I_{sc} \propto$  irradiance;  $V_{oc}$  varies weakly (logarithmically) with irradiance and temperature.

**Solution.** Using a single-diode model  $I(V) = I_{sc}[1 - e^{(V-V_{oc})/V_t}]$ , the condition  $dP/dV = 0$  gives

$$1 = e^{(V_{mp}-V_{oc})/V_t} \left( 1 + \frac{V_{mp}}{V_t} \right).$$

For sail  $\alpha$  this is satisfied at  $V_{mp} \approx 16.26$  V, where  $I \approx 7.33$  A and

$$P_{max} = V_{mp} I(V_{mp}) \approx 16.26 \times 7.33 \approx 119.3 \text{ W.}$$

This matches the engineering rule of thumb  $V_{mp} \approx 0.81 V_{oc}$ . For the remaining sails the peak is found the same way;  $I_{sc}$  scales with irradiance ( $\beta$ :  $420$  W/m<sup>2</sup>  $\Rightarrow I_{sc} \approx 3.36$  A;  $\gamma$ :  $1400$  W/m<sup>2</sup>  $\Rightarrow I_{sc} \approx 11.2$  A), so each curve’s peak shifts but stays near  $V \approx 0.8 V_{oc}$ .

### INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW

Drag the tracker dot to the **peak of the gold  $P$ - $V$  curve**:

Sail  $\alpha$ :  $V_{mp} \approx 16.26 \text{ V}$ ,  $P_{max} \approx 119.3 \text{ W}$  (100 % of  $P_{max}$ ). Press **LOCK MPP & HARVEST**.

For each subsequent sail, drag to that curve's peak ( $V_{mp} \approx 0.8 V_{oc}$ ).

## 3 Task 3 — Mohr's Circle Audit — Principal Stresses

**Problem.** A stress element with components  $(\sigma_x, \sigma_y, \tau_{xy})$  must be analysed by constructing Mohr's circle to expose the principal stresses  $\sigma_1, \sigma_2$ . (In the screenshot the structural scanner is offline; the procedure below applies once **SCAN** reveals the element values.)

### Governing relations

$$C = \frac{\sigma_x + \sigma_y}{2}, \quad R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2},$$

$$\sigma_1 = C + R, \quad \sigma_2 = C - R, \quad \tau_{max} = R.$$

**Solution (worked template).** After pressing SCAN, read  $\sigma_x, \sigma_y, \tau_{xy}$  from the top-right stress element. For example, with the representative values  $\sigma_x = 120 \text{ MPa}$ ,  $\sigma_y = 40 \text{ MPa}$ ,  $\tau_{xy} = 30 \text{ MPa}$ :

$$C = \frac{120 + 40}{2} = 80 \text{ MPa}, \quad R = \sqrt{40^2 + 30^2} = 50 \text{ MPa},$$

$$\sigma_1 = 80 + 50 = 130 \text{ MPa}, \quad \sigma_2 = 80 - 50 = 30 \text{ MPa}, \quad \tau_{max} = 50 \text{ MPa}.$$

The circle is centred on the  $\sigma$ -axis at  $(C, 0)$  with radius  $R$ ; it must pass through both diamond points  $A(\sigma_x, +\tau_{xy})$  and  $B(\sigma_y, -\tau_{xy})$ .

### INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW

1. Drag the resize handle until the orange circle radius equals  $R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$  (circle passes through both blue/purple diamonds).
2. Click the  $\sigma_1$  **node** (right intercept,  $\sigma_1 = C + R$ ) and the  $\sigma_2$  **node** (left intercept,  $\sigma_2 = C - R$ ).
3. Press **COMPLETE CHECKLIST**.

Template example ( $\sigma_x=120, \sigma_y=40, \tau_{xy}=30$ ):  $R = 50$ ,  $\sigma_1 = 130 \text{ MPa}$ ,  $\sigma_2 = 30 \text{ MPa}$ .

## 4 Task 4 — Radar Convolution — Cross-Correlation Peak

**Problem.** A radar echo  $x(t)$  contains three blips; only one is the real return (the others are decoy pods). Sliding a search pulse  $h(t)$  over the echo, the real target produces the *global maximum* of the cross-correlation.

### Governing relations

$$R(\tau) = \sum_t h(t) x(t + \tau), \quad \tau^* = \arg \max_{\tau} R(\tau).$$

$h(t)$  = search pulse,  $x(t)$  = radar echo.

**Solution.** This is a matched-filter detection problem. Cross-correlation measures similarity between the pulse and the echo at every lag  $\tau$ . The *real* wizard return has the same shape as the search

pulse, so the correlation reaches its highest peak there; decoys, being only partially similar, yield smaller secondary peaks. The optimal lag  $\tau^*$  is the abscissa of the tallest peak of  $R(\tau)$  — on the cross-correlation panel this is the dominant central peak, clearly higher than the two flanking (decoy) peaks.

#### INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW

Drag the amber search pulse so the cyan correlation dot climbs to the **tallest (global-maximum) peak** of  $R(\tau)$  — the dominant central lobe, not the lower side peaks. Stop within  $\pm 10$  samples of that peak (dot turns green) and press **LOCK TARGET**.

## 5 Task 5 — Arcane Capacitor — Displacement-Current B-Field

**Problem.** A time-varying  $E$ -field in a capacitor gap produces a displacement current and hence a magnetic field. Tune the AC frequency so the rim field  $B_{rim}$  matches the target.

Live data:  $f = 69.0$  MHz,  $B_{rim} = 127,952.0$  nT,  $B_{target} = 195,896.8$  nT, error = 34.7 %.

#### Governing relations

$$I_d = \varepsilon_0 A (2\pi f) E_0, \quad B = \frac{\mu_0 I_d}{2\pi r} \implies B \propto f.$$

**Solution.** Because every other quantity is fixed, the rim field is *linearly proportional to frequency*,  $B = kf$  with  $k = B_{rim}/f = 127,952.0/69.0 = 1854.4$  nT/MHz. The frequency that hits the target is

$$f_{target} = f \cdot \frac{B_{target}}{B_{rim}} = 69.0 \times \frac{195,896.8}{127,952.0} = 69.0 \times 1.5310 = 105.6 \text{ MHz}.$$

This lies well inside the  $\pm 5\%$  window ( $B$  between 186.1 and 205.7  $\mu$ T  $\Rightarrow f \in [100.4, 110.9]$  MHz).

#### INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW

Drag the frequency slider to  $f \approx 105.6$  **MHz**  
(acceptable range 100.4–110.9 MHz for  $\pm 5\%$ ). Press **LOCK ON**.

## 6 Task 6 — Counter-Flow Heat Exchanger — NTU- $\varepsilon$ Method

**Problem.** Cool the reactor coolant outlet below 75°C. Inlets:  $T_{h,in} = 200^\circ\text{C}$ ,  $T_{c,in} = 20^\circ\text{C}$ ;  $UA = 5$  kW/K,  $c_p = 4.18$  kJ/(kg·K). Adjust the hot- and cold-side mass flow rates  $\dot{m}_h, \dot{m}_c$ .

#### Governing relations

$$C = \dot{m}c_p, \quad \text{NTU} = \frac{UA}{C_{min}}, \quad C_r = \frac{C_{min}}{C_{max}},$$

$$\varepsilon = \frac{1 - e^{-\text{NTU}(1-C_r)}}{1 - C_r e^{-\text{NTU}(1-C_r)}} \quad (\text{counter-flow}),$$

$$Q = \varepsilon C_{min} (T_{h,in} - T_{c,in}), \quad T_{h,out} = T_{h,in} - \frac{Q}{C_h}.$$

**Solution.** Lowering  $\dot{m}_h$  reduces  $C_h$ : this both raises NTU ( $= UA/C_{min}$ ) and lowers  $C_r$ , and both effects increase  $\varepsilon$  and the hot-side temperature drop. Raising  $\dot{m}_c$  further lowers  $C_r$ . With the hot side as  $C_{min}$ ,  $T_{h,out} = T_{h,in} - \varepsilon \Delta T_{max}$ , so the win needs  $\varepsilon \geq 125/180 = 0.694$ .

Taking  $\dot{m}_h = 0.80 \text{ kg/s}$  and  $\dot{m}_c = 2.50 \text{ kg/s}$ :

$$C_h = 3.34, \quad C_c = 10.45 \text{ kW/K}, \quad C_r = 0.320, \quad \text{NTU} = \frac{5}{3.34} = 1.50,$$

$$\varepsilon = \frac{1 - e^{-1.50(0.68)}}{1 - 0.320 e^{-1.50(0.68)}} = 0.722, \quad Q = 0.722 \times 3.34 \times 180 = 434 \text{ kW},$$

$$T_{h,out} = 200 - \frac{434}{3.34} = 70.1^\circ\text{C} < 75^\circ\text{C}. \checkmark$$

(The exact boundary  $T_{h,out} = 75^\circ\text{C}$  occurs at  $\dot{m}_h \approx 0.86 \text{ kg/s}$ ;  $0.80 \text{ kg/s}$  gives a safe margin.)

#### INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW

Set the hot-flow slider to  $\dot{m}_h \approx 0.80 \text{ kg/s}$  (toward minimum) and the cold-flow slider to  $\dot{m}_c \approx 2.50 \text{ kg/s}$  (toward maximum).

Result:  $\varepsilon \approx 72\%$ ,  $T_{h,out} \approx 70^\circ\text{C}$ . Press **TRIGGER SHUTDOWN**.

## 7 Task 7 — Sentry-Bot MK-VII — 3-Joint Forward Kinematics

**Problem.** Place the end-effector of a 3-joint arm ( $\theta_1$  yaw,  $\theta_2$  shoulder,  $\theta_3$  elbow) within 12 units of the Obsidian Lock at  $(x, y, z) = (167.01, 50.41, -16.23)$ .

#### Governing relations

$$x = \cos \theta_1 \cdot r, \quad y = \sin \theta_1 \cdot r, \quad r = L_1 \cos \theta_2 + L_2 \cos(\theta_2 + \theta_3),$$

$$z = L_1 \sin \theta_2 + L_2 \sin(\theta_2 + \theta_3).$$

**Solution.** *Calibration.* The displayed pose  $\theta_1 = 16.6^\circ$ ,  $\theta_2 = -3.1^\circ$ ,  $\theta_3 = -4.5^\circ$  yields  $EE = (186.0, 55.5, -17.3)$ . From the planar pair of equations this fixes the link lengths  $L_1 \approx 108.6$ ,  $L_2 \approx 86.4$  units.

*Inverse kinematics for the target.*

$$\theta_1 = \text{atan2}(50.41, 167.01) = 16.8^\circ,$$

$$r_t = \sqrt{167.01^2 + 50.41^2} = 174.5, \quad R_t = \sqrt{r_t^2 + z_t^2} = 175.2,$$

$$\cos \theta_3 = \frac{R_t^2 - L_1^2 - L_2^2}{2L_1L_2} = 0.609 \Rightarrow \theta_3 = -52.5^\circ,$$

$$\theta_2 = \text{atan2}(z_t, r_t) + \arccos \frac{L_1^2 + R_t^2 - L_2^2}{2L_1R_t} = 17.7^\circ.$$

Forward-checking these angles returns  $EE \approx (167.0, 50.4, -16.2)$  — essentially on target (dist  $\approx 0.1 \ll 12$ ).

#### INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW

Set the canvas sliders to:

$\theta_1 \approx 16.8^\circ$  (**yaw**),  $\theta_2 \approx 17.7^\circ$  (**shoulder**),  $\theta_3 \approx -52.5^\circ$  (**elbow**).

The elbow must fold in sharply (reach drops  $195 \rightarrow 175$ ). Press **UNLOCK DOOR**.

## 8 Task 8 — Rankine Cycle — Thermal Efficiency

**Problem.** Engineer a Rankine cycle exceeding 32% thermal efficiency by adjusting boiler temperature  $T_H$  and condenser temperature  $T_L$ . Start:  $T_H = 180^\circ\text{C}$ ,  $T_L = 90^\circ\text{C}$ ,  $\eta = 18\%$ .

**Governing relations**

$$\eta = \frac{W_{net}}{Q_{in}}, \quad \eta_{Carnot} = 1 - \frac{T_L}{T_H} \quad (\text{absolute temperatures}).$$

**Solution.** Raising  $T_H$  and lowering  $T_L$  enlarges the enclosed area of the  $T$ - $s$  cycle, which is the net work  $W_{net}$ , while the Carnot ceiling rises. The current cycle ( $\eta = 18\%$ ) sits close to its Carnot bound  $1 - 363/453 = 19.9\%$ . Driving the bars to their extremes  $T_H = 350^\circ\text{C}$  (623 K) and  $T_L = 30^\circ\text{C}$  (303 K):

$$\eta_{Carnot} = 1 - \frac{303}{623} = 51.4\%.$$

Maintaining the same fraction of the Carnot limit ( $\eta/\eta_{Carnot} \approx 0.9$ ) gives an actual cycle efficiency  $\eta \approx 0.9 \times 51.4 \approx 46\%$  — comfortably above the 32% target.

**INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW**

Drag the **red  $T_H$  bar up to  $\approx 350^\circ\text{C}$**  and the **blue  $T_L$  bar down to  $\approx 30^\circ\text{C}$** .  
Result:  $\eta_{Carnot} = 51.4\%$ , cycle  $\eta \approx 45\text{--}46\% > 32\%$ . Press **ACTIVATE OVERDRIVE**.

**9 Task 9 — Nyquist Hack — Encirclement Count**

**Problem.** For Shield Alpha with open-loop transfer function  $G(s) = \frac{1}{(s+1)^2}$ , count the clockwise encirclements  $N$  of the critical point  $(-1, 0)$  and determine stability.

**Governing relations**

$$Z = N + P, \quad \begin{cases} N = \# \text{ clockwise encirclements of } (-1, 0) \\ P = \# \text{ open-loop unstable poles} \\ Z > 0 \Rightarrow \text{UNSTABLE} \end{cases}$$

**Solution.**  $G(s) = 1/(s+1)^2$  has a double pole at  $s = -1$ , in the left-half plane, so  $P = 0$ . The Nyquist locus starts at  $G(j0) = 1 (= 1\angle 0^\circ)$  and spirals into the origin as  $\omega \rightarrow \infty$ , with phase sweeping only from  $0^\circ$  to  $-180^\circ$ . The plotted curve is a small loop confined near the origin and the point  $(+1, 0)$ ; it never wraps around the critical point  $(-1, 0)$ . Hence

$$N = 0, \quad Z = N + P = 0 \quad \Rightarrow \quad \text{closed-loop STABLE.}$$

**INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW**

Select  $N = 0$ .  
(The locus does not encircle  $(-1, 0)$ ; with  $P = 0$ ,  $Z = N + P = 0$  — Shield Alpha is stable.)

**10 Task 10 — Torsional Stress — Yield Torque of a Shaft**

**Problem.** Find the exact torque  $T_{max}$  that brings a solid vault shaft to its yield point at the outer fibre.

Data:  $R = 53\text{ mm}$ ,  $J = 12.394 \times 10^{-6}\text{ m}^4$ ,  $\tau_{yield} = 130\text{ MPa}$ .

### Governing relations

$$\tau(r) = \frac{T r}{J}, \quad J = \frac{\pi R^4}{2}, \quad \tau_{max} = \frac{T R}{J} \Rightarrow T_{max} = \frac{\tau_{yield} J}{R}.$$

**Solution.** Shear stress varies linearly with radius (zero at the centre, maximum at the surface); only the outer fibre reaches yield first. Setting  $\tau_{max} = \tau_{yield}$ :

$$T_{max} = \frac{\tau_{yield} J}{R} = \frac{(130 \times 10^6)(12.394 \times 10^{-6})}{0.053} = 30,400 \text{ N} \cdot \text{m} \approx 30.40 \text{ kN} \cdot \text{m}.$$

*Check:* at  $T = 29.992 \text{ kN} \cdot \text{m}$  the model reads  $\tau_{max} = 128 \text{ MPa} = 99\%$  of  $\tau_{yield}$  — consistent.

### INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW

Drag the torque slider to  $T_{max} = 30.40 \text{ kN} \cdot \text{m}$  ( $\approx 30,400 \text{ N} \cdot \text{m}$ ).  
The green dot meets the red yield line at  $r = R$  ( $\tau_{max} = 130 \text{ MPa}$ ). Press **APPLY T\_MAX**.

## 11 Task 11 — Void Frequency Tuner — RLC Resonance

**Problem.** An RLC filter must be tuned by sliding the inductor knob  $L$  so the receiver's resonant frequency  $f_0$  aligns with the wizard's spike  $f_T$ . Data:  $C = 1.0 \mu\text{F}$ ,  $Q = 12$ , current  $L = 3.57 \text{ mH}$  giving  $f_0 = 2663 \text{ Hz}$  at 100 % signal.

### Governing relations

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \quad \Rightarrow \quad L = \frac{1}{(2\pi f_T)^2 C}.$$

**Solution.** Resonance occurs where inductive and capacitive reactances cancel. To shift the green resonant peak onto the red target spike, set  $L$  from the inverted resonance formula. At the displayed setting

$$f_0 = \frac{1}{2\pi\sqrt{(3.57 \times 10^{-3})(1.0 \times 10^{-6})}} = 2664 \text{ Hz},$$

which already coincides with  $f_T$  — the receiver reads SIGNAL 100 %, so  $f_T \approx 2663 \text{ Hz}$  and  $L = 3.57 \text{ mH}$  is the correct inductance.

### INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW

Set the inductor knob to  $L = 3.57 \text{ mH} \Rightarrow f_0 = 2663 \text{ Hz} = f_T$  (Signal 100 %). Press **LOCK SIGNAL**.

General retune:  $L = \frac{1}{(2\pi f_T)^2 C}$ .

## 12 Task 12 — Beam Deflection — Location of Maximum Deflection

**Problem.** A simply supported bridge beam ( $L = 6.00 \text{ m}$ ,  $R_a = 29.5 \text{ kN}$ ,  $R_b = 42.1 \text{ kN}$ ) carries three point loads. Predict the position of maximum deflection (tolerance  $\pm 0.55 \text{ m}$ ).



### Governing relations

$$EI y'' = M(x), \quad \delta_{max} \text{ where } y'(x) = 0,$$
$$V(x) = 0 \Rightarrow dM/dx = 0 \Rightarrow M(x) \text{ is maximum.}$$

**Solution.** For a simply supported beam the elastic curve is flattest (zero slope, maximum deflection) very close to the section of maximum bending moment. The bending-moment diagram peaks at  $x \approx 3.06$  m (where the shear  $V(x)$  changes sign under the dominant 31 kN load). Hence the maximum deflection occurs at

$$x_{\delta_{max}} \approx 3.06 \text{ m,}$$

essentially mid-span ( $L/2 = 3.00$  m), comfortably within the  $\pm 0.55$  m tolerance band.

### INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW

Drag the amber marker to  $x \approx 3.06$  m (the BMD peak; 3.00 m mid-span is also within tolerance). Press **REVEAL DEFLECTION**.

## 13 Task 13 — Singularity Trap — Jacobian Determinant

**Problem.** A 2-link arm must reach the gold target while keeping the normalised Jacobian determinant above threshold to avoid a kinematic singularity.

### Governing relations

$$\det(\mathbf{J}) = L_1 L_2 \sin \theta_2, \quad \frac{|\det(\mathbf{J})|}{L_1 L_2} = |\sin \theta_2| > 0.25.$$
$$\theta_2 = 0^\circ \text{ (fully extended) or } \pm 180^\circ \text{ (fully folded)} \Rightarrow \text{singularity.}$$

**Solution.** The Jacobian determinant collapses to zero when the arm is straight or fully folded, where the manipulator loses a degree of freedom. The safety condition  $|\sin \theta_2| > 0.25$  requires

$$14.5^\circ < \theta_2 < 165.5^\circ.$$

Manipulability is maximised at  $\theta_2 = 90^\circ$  ( $|\det|/L_1 L_2 = 1$ ). Reaching the target with the elbow bent near  $\theta_2 \approx 87^\circ$  (as in the displayed pose,  $\theta_1 = 8.5^\circ$ ) keeps  $|\det|/L_1 L_2 \approx 0.998$  — far from any singularity zone.

### INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW

Drag the Elbow/EE handles so the arm reaches the gold target with the elbow well-bent:  $\theta_2 \approx 87^\circ$  (keep  $14.5^\circ < \theta_2 < 165.5^\circ$ ;  $90^\circ$  is ideal),  $\theta_1 \approx 8.5^\circ$ .  
Result: dist < 28 px and  $|\det \mathbf{J}|/L_1 L_2 \approx 1$ . Press **ESCAPE!**.

## 14 Task 14 — PID Stabilizer — Tuning the Wind-Void Crossing

**Problem.** Tune the PID gains  $K_p, K_i, K_d$  to reduce the oscillation energy below 1250. Start:  $K_p = 0.50$ ,  $K_i = 0.10$ ,  $K_d = 0.30$ ,  $E = 1274.1$ .

### Governing relations

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de}{dt}.$$

$K_p$ : reacts to current error;  $K_i$ : removes accumulated drift;  $K_d$ : damps oscillation by anticipating trends.

**Solution.** The flight path shows a lightly damped oscillation, and the energy ( $E = 1274$ ) is only marginally above the 1250 target. The derivative term is the one that suppresses oscillation, so it should be increased; the proportional term is slightly too aggressive (it feeds the overshoot) and is trimmed; the integral term is kept small to avoid integral wind-up that would re-introduce slow oscillation:

$$K_d \uparrow, \quad K_p \downarrow \text{ slightly}, \quad K_i \downarrow.$$

A balanced set near  $K_p \approx 0.40$ ,  $K_i \approx 0.05$ ,  $K_d \approx 0.55$  adds damping without making the craft sluggish, pulling  $E$  below 1250.

**INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW**

Set the dials to approximately  $K_p \approx 0.40$ ,  $K_i \approx 0.05$ ,  $K_d \approx 0.55$   
(increase  $K_d$  to damp oscillation, ease  $K_p$  down, keep  $K_i$  small). Fine-tune until OSCILLATION ENERGY < 1250, then press **ATTEMPT CROSSING**.

## 15 Task 15 — Mach Shock — Normal Shock in a C-D Nozzle

**Problem.** Position a normal shock in the diverging section of a converging–diverging nozzle so the exit pressure ratio hits  $P_{exit}/P_0 = 0.738$  ( $\pm 4\%$ ). Data:  $\gamma = 1.4$ , exit area ratio  $A_e/A^* = 2.125$ , shock at  $x > 0.5$ .

**Governing relations**

$$\frac{A}{A^*} = \frac{1}{M} \left[ \frac{2}{\gamma + 1} \left( 1 + \frac{\gamma - 1}{2} M^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}},$$

$$M_2^2 = \frac{M_1^2 + 5}{7M_1^2 - 1}, \quad \frac{P_{0,2}}{P_{0,1}} \text{ (normal-shock stagnation loss).}$$

**Solution.** Moving the shock downstream raises the pre-shock Mach number  $M_1$ , strengthening the shock and increasing the stagnation-pressure loss; the subsonic flow after the shock then expands to a different exit pressure. Sweeping the shock area ratio  $A_s/A^*$ :

$$\frac{A_s}{A^*} = 1.40 \Rightarrow \frac{P_{exit}}{P_0} = 0.760, \quad \frac{A_s}{A^*} = 1.50 \Rightarrow 0.715,$$

so the target 0.738 is reached at  $A_s/A^* \approx 1.445$  ( $M_1 \approx 1.81$ ). With the fitted nozzle geometry  $A/A^* = 1 + 4.5(x - 0.5)^2$ , this corresponds to

$$x = 0.5 + \sqrt{\frac{1.445 - 1}{4.5}} \approx 0.81.$$

The full normal-shock calculation at this station returns  $P_{exit}/P_0 \approx 0.740$  — inside the  $\pm 4\%$  window.

**INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW**

Drag the shock marker into the diverging section to  $x \approx 0.81$  (area ratio  $A_s/A^* \approx 1.45$ ).  
Result:  $P_{exit}/P_0 \approx 0.738$ . Press **FIRE HACK**.

## 16 Task 16 — Kalman Teleportation Tracker — Optimal Estimate

**Problem.** Each round, noisy yellow sensor blips appear; click where the teleporting wizard truly is. The Kalman filter then fuses the estimate. Parameters:  $Q = 8$  (process noise),  $R = 15$  (sensor noise), initial  $P = 50$ .

### Governing relations

Predict:  $\hat{x}_k^- = \hat{x}_{k-1}$ ,  $P_k^- = P_{k-1} + Q$ .

Update:  $K_k = \frac{P_k^-}{P_k^- + R}$ ,  $\hat{x}_k = \hat{x}_k^- + K_k(z_k - \hat{x}_k^-)$ ,  $P_k = (1 - K_k)P_k^-$ .

**Solution.** On the first round the prior is highly uncertain ( $P = 50 \gg R$ ), so the Kalman gain is large ( $K \approx 50/65 \approx 0.77$ ) and the estimate is dominated by the measurements. With several independent blips the optimal estimate is their *average* (the maximum-likelihood location of the source). Therefore click at the **centroid of the yellow blip cluster**. As rounds progress,  $P$  shrinks,  $K$  falls, and the estimate increasingly trusts the converged prior — but the click rule (centre of the blip pattern, nudged toward the running estimate) stays the same.

### INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW

Click on the grid at the **centroid (average position) of the visible yellow blips** — for Round 1 the blips cluster near  $x \approx 33$ – $38$ , so click at  $x \approx 35$ .  
Each round: click the blip-cluster centre; the filter converges ( $P \downarrow$ ) over the 7 rounds.

## 17 Task 17 — Spectral De-Scrambler — FFT Carrier Frequencies

**Problem.** The raw audio is white noise in the time domain; switching to the frequency domain (FFT) reveals the wizard's jamming spikes. Click the three sharp resonant peaks to lock the carrier frequencies.

### Governing relations

$$X(k) = \sum_n x(n) e^{-i 2\pi k n / N}.$$

$|X(k)|$  isolates the amplitude of periodic sine components buried in random noise  $x(n)$ .

**Solution.** A Fourier transform decomposes the time signal into its constituent frequencies. Random noise spreads energy thinly across all bins, while a periodic carrier concentrates energy into a single sharp spike. The FFT analyzer shows three such resonant peaks; reading their abscissae gives the three hidden carrier frequencies:

$$f_1 \approx 45 \text{ Hz}, \quad f_2 \approx 212 \text{ Hz}, \quad f_3 \approx 395 \text{ Hz}.$$

### INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW

Click the three sharp peaks in the FFT panel at  $f_1 \approx 45 \text{ Hz}$ ,  $f_2 \approx 212 \text{ Hz}$ ,  $f_3 \approx 395 \text{ Hz}$  (CH.1, CH.2, CH.3). Then press **DECODE**.

## 18 Task 18 — The Prism Shield — LQR Controller Q/R Balance

**Problem.** Select the LQR weighting on the state-space plot to balance fast stabilisation against efficient energy use.

### Governing relations

$$J = \int_0^\infty (\mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{u}^\top \mathbf{R} \mathbf{u}) dt, \quad \dot{\mathbf{x}} = \mathbf{A} \mathbf{x} + \mathbf{B} \mathbf{u}.$$

$\mathbf{Q}$  penalises state error (speed);  $\mathbf{R}$  penalises control effort (energy).

**Solution.** The LQR cost trades two competing objectives. A large  $\mathbf{Q}$  (relative to  $\mathbf{R}$ ) drives the state to zero quickly but demands aggressive, energy-hungry control; a large  $\mathbf{R}$  saves energy but stabilises slowly. The mission requires the platform to settle *before the shield rotates* yet still *retain energy for the final blast* — so neither extreme is acceptable. The sweet spot is a roughly balanced ratio,  $\mathbf{Q}/\mathbf{R} \approx 1$  (with a mild bias toward  $\mathbf{Q}$  for prompt settling). On the Position–Velocity selector this is a point near the centre, slightly into the high- $\mathbf{Q}$  region, giving a well-damped trajectory that spirals smoothly into the origin without overshoot.

### INPUT TO ENTER / ACTION IN INTERACTIVE WINDOW

Click the state-space plot at a **balanced point** — moderate  $\mathbf{Q}$  and moderate  $\mathbf{R}$  ( $\mathbf{Q}/\mathbf{R} \approx 1$ , slightly biased toward higher  $\mathbf{Q}$ ): near the plot centre, just into the high- $\mathbf{Q}$  side. Press **TEST CONFIG**.

(Avoid the extremes: pure high- $\mathbf{Q}$  = overshoot/energy waste; pure high- $\mathbf{R}$  = too slow.)

## 19 Summary of All Inputs

| #  | Task                 | Value to enter / action                                                                                         |
|----|----------------------|-----------------------------------------------------------------------------------------------------------------|
| 1  | Column Stability     | Brace at $a = 2.00$ m; $P_{cr} = 197.4$ kN.                                                                     |
| 2  | Solar Sail MPPT      | Tracker at $V_{mp} \approx 16.26$ V, $P \approx 119.3$ W (sail $\alpha$ ).                                      |
| 3  | Mohr's Circle        | $R = \sqrt{((\sigma_x - \sigma_y)/2)^2 + \tau_{xy}^2}$ ; click $\sigma_1 = C + R$ , $\sigma_2 = C - R$ .        |
| 4  | Radar Convolution    | Drag pulse to the global-maximum (tallest) correlation peak.                                                    |
| 5  | Arcane Capacitor     | $f \approx 105.6$ MHz.                                                                                          |
| 6  | Counter-Flow HX      | $\dot{m}_h \approx 0.80$ kg/s, $\dot{m}_c \approx 2.50$ kg/s $\Rightarrow T_{h,out} \approx 70^\circ\text{C}$ . |
| 7  | Sentry-Bot FK        | $\theta_1 \approx 16.8^\circ$ , $\theta_2 \approx 17.7^\circ$ , $\theta_3 \approx -52.5^\circ$ .                |
| 8  | Rankine Cycle        | $T_H \approx 350^\circ\text{C}$ , $T_L \approx 30^\circ\text{C} \Rightarrow \eta \approx 45\%$ .                |
| 9  | Nyquist Hack         | $N = 0$ (stable; $Z = N + P = 0$ ).                                                                             |
| 10 | Torsional Stress     | $T_{max} = 30.40$ kN·m.                                                                                         |
| 11 | Void Frequency Tuner | $L = 3.57$ mH $\Rightarrow f_0 = 2663$ Hz.                                                                      |
| 12 | Beam Deflection      | Marker at $x \approx 3.06$ m.                                                                                   |
| 13 | Singularity Trap     | $\theta_2 \approx 87^\circ$ (keep $14.5^\circ < \theta_2 < 165.5^\circ$ ).                                      |
| 14 | PID Stabilizer       | $K_p \approx 0.40$ , $K_i \approx 0.05$ , $K_d \approx 0.55$ .                                                  |
| 15 | Mach Shock           | Shock at $x \approx 0.81$ ( $A_s/A^* \approx 1.45$ ).                                                           |
| 16 | Kalman Tracker       | Click blip-cluster centroid ( $\approx x = 35$ , Round 1).                                                      |

| #  | Task                  | Value to enter / action                                                                   |
|----|-----------------------|-------------------------------------------------------------------------------------------|
| 17 | Spectral De-Scrambler | Peaks at 45, 212, 395 Hz.                                                                 |
| 18 | Prism Shield (LQR)    | Balanced $\mathbf{Q}/\mathbf{R} \approx 1$ , centre of plot (slight high- $\mathbf{Q}$ ). |

*Notes.* Tasks 3, 4, 16 and 18 depend on data that is hidden or graphical in the live page (offline scanner, correlation trace, sensor blips, empty selector); for these the governing method is given so the exact on-screen value can be applied directly. All numeric answers were verified against the live readouts shown in the task screenshots.